

Total No. of Questions : 8]

SEAT No. :

PD-5303

[Total No. of Pages : 2

[6403]-43

T.E. (Computer Engineering)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - VI) (310251)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is allowed.

Q1) a) What is Model Building elaborate this phase of data analytics with the help of a suitable example? **[8]**

b) List out different stakeholders of an analytics project. What do they usually expect at the conclusion (key outputs) of a project? **[9]**

OR

Q2) a) Explain Descriptive, Diagnostic, Predictive analytics. **[8]**

b) List and explain the various activities involved in identifying potential data resources as a part of discovery phase in Data Analytics Life Cycle? **[9]**

Q3) a) What is association rule mining? Describe the working of the Apriori algorithm with an example. **[9]**

b) Explain how decision trees are constructed using information gain and entropy. Illustrate with a small example. **[9]**

OR

Q4) a) Explain Naïve Bayes' classifier and its applications. **[9]**

b) Consider a dataset with binary classes and two features: "Loan Amount" and "Default History." Show how logistic regression could be applied for loan default prediction. **[9]**

P.T.O.

- Q5) a)** Explain the holdout method. Differentiate training set, validation set, and test set. [8]
- b)** Given the confusion matrix below. Calculate Accuracy, Precision, Recall and F1-score. [9]

	Predicted Yes	Predicted No
Actual Yes	70	30
Actual No	20	80

OR

- Q6) a)** Explain the following Text Analysis steps with suitable example [8]
- Part-of-speech (POS) tagging
 - Lemmatization
- b)** Use K-Means Clustering for the following points and determine the centroids after one iteration. Assume initial centroids as A(1,1), B(5,7). Points: (1,2), (2,1), (3,5), (6,8), (7,6), (5,5) [9]

- Q7) a)** Explain Hadoop Architecture with a neat diagram. Highlight the roles of NameNode and DataNode [9]
- b)** Compare Tableau, Power BI, and Matplotlib for data visualization. Discuss scenarios where each tool is best suited. [9]

OR

- Q8) a)** What is Data Visualization? Describe the challenges of data visualization. [9]
- b)** Write short notes on the following : [9]
- Map Reduce
 - HDFS
 - Hive

